

$$MI(x_k) = \sum_{k=1}^t \left(b_1 \cdot I^* \cdot \frac{x_k^2}{X \cdot v_k} \right) + (1 - b_1) \cdot I^* \quad (7.15)$$

$$\text{with } \sum x_k = X$$

The derivation of the trade schedule market impact formula above (Equation 7.15) is:

Start with the instantaneous cost estimate I^* . This value is allocated to each trade period based on the percentage of the order transacted in that period. If x_k shares of the total order X were executed in period k then the percentage I^* allocated to period k is:

$$I^* \cdot \frac{x_k}{X}$$

Therefore, the percentage of temporary impact allocated to period k is $b_1 \cdot I^* \cdot \frac{x_k}{X}$ and the percentage of permanent impact allocated to period k is $(1 - b_1) \cdot I^* \cdot \frac{x_k}{X}$.

The temporary impact cost is allocated to the investor based on the percentage of volume of the trade in that period. This is:

$$b_1 \cdot I^* \cdot \frac{x_k}{X} \cdot \frac{x_k}{x_k + v_k}$$

For simplicity, however, we rewrite temporary impact cost in terms of the trade rate as follows:

$$b_1 \cdot I^* \cdot \frac{x_k}{X} \cdot \frac{x_k}{v_k}$$

Finally, the total market impact cost of a trade schedule over all periods is determined by summing the cost over all periods. That is:

$$MI(x_k) = \sum_{k=1}^n b_1 \cdot I^* \cdot \frac{x_k}{X} \cdot \frac{x_k}{v_k} + \sum_{k=1}^n \frac{x_k}{X} \cdot (1 - b_1) \cdot I^*$$

This formulation is then simplified as:

$$MI(x_k) = \sum_{k=1}^n \left(b_1 \cdot I^* \cdot \frac{x_k^2}{X \cdot v_k} \right) + (1 - b_1) \cdot I^*$$

The units of the market impact cost formula will be the same as the units used in the I^* formula shown in Equations 7.10 to 7.12. Kissell and Glantz (2003) and Kissell, Glantz, and Malamut (2004) provide alternative derivations of the trade schedule market impact formulation.